

Central Limit Theorem

MDragon Data Tools - Study Guide

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1. What is the Central Limit Theorem (CLT)?

The **Central Limit Theorem** states that:

When you take sufficiently large random samples from any population (with any shape), the distribution of sample means will be approximately normally distributed, regardless of the population's original distribution.

In Simple Terms

- Take many samples from a population
- Calculate the mean of each sample
- Plot all those sample means

- The distribution will look like a bell curve (normal distribution)
 - This works even if the original population is NOT normal!
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2. Why It Matters

The CLT is one of the most important theorems in statistics because:

1. **Enables inference** - We can make conclusions about populations using normal distribution theory
2. **Works with any distribution** - Original population can be skewed, uniform, bimodal, etc.
3. **Foundation for hypothesis testing** - Allows us to test claims about population means
4. **Basis for confidence intervals** - Enables us to estimate population parameters
5. **Justifies using Z and T tests** - Even when population isn't normal

Real-World Applications

- Political polling and surveys
 - Quality control in manufacturing
 - Medical research and clinical trials
 - Financial risk assessment
 - A/B testing in business
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3. Key Concepts

Sampling Distribution of Sample Means

- **Population:** All individuals of interest
- **Sample:** Subset taken from population
- **Sample mean ($\bar{x}$):** Average of sample values
- **Sampling distribution:** Distribution of ALL possible sample means

Properties of the Sampling Distribution

1. Mean of sampling distribution:

$$\mu_{\bar{x}} = \mu \text{ (equals population mean)}$$

2. Standard deviation of sampling distribution (Standard Error):

$$\sigma_{\bar{x}} = \sigma / \sqrt{n}$$

where:

σ = population standard deviation

n = sample size

3. Shape:

- Approximately normal (if $n \geq 30$ or population is normal)
- More normal as n increases
- Center closer to μ as n increases
- Spread decreases as n increases

4. Requirements and Conditions

For CLT to Apply:

1. Sample Size Rule of Thumb:

- $n \geq 30$: CLT applies for most distributions
- $n < 30$: CLT still works if population is approximately normal
- **Severely skewed**: May need $n > 30$ (sometimes $n \geq 40$)

2. Independence:

- Random sampling

- If sampling without replacement: $n \leq 0.05N$ (sample < 5% of population)

3. Sample vs Population:

- Each sample is a random, independent observation
- Samples drawn from the same population

5. Step-by-Step Examples

Example 1: Basic CLT Application

 **Problem:** A population of test scores has $\mu = 70$ and $\sigma = 15$. If we take samples of size $n = 36$, describe the sampling distribution of $\bar{x}$.

 **Solution:**

1. Check if CLT applies:

- $n = 36 \geq 30$ ✓
- Random sampling assumed ✓

2. Characteristics of sampling distribution:

Shape: Approximately normal (due to CLT)

Mean: $\mu_{\bar{x}} = \mu = 70$

Standard Error: $\sigma_{\bar{x}} = \sigma / \sqrt{n}$
 $= 15 / \sqrt{36}$
 $= 15 / 6$
 $= 2.5$

 **Answer:**

- The sampling distribution of $\bar{x}$ is approximately normal
- Center: $\mu_{\bar{x}} = 70$
- Spread: $\sigma_{\bar{x}} = 2.5$

Interpretation: Sample means will typically be within 2.5 points of the true mean (70).

Example 2: Probability About Sample Mean

 **Problem:** Using the scenario from Example 1, what is the probability that a random sample of 36 scores has a mean greater than 75?

 **Solution:**

1. We know from Example 1:

- $\mu_{\bar{x}} = 70$
- $\sigma_{\bar{x}} = 2.5$
- Distribution is approximately normal

2. Calculate z-score:

$$\begin{aligned} z &= (\bar{x} - \mu_{\bar{x}}) / \sigma_{\bar{x}} \\ &= (75 - 70) / 2.5 \\ &= 5 / 2.5 \\ &= 2 \end{aligned}$$

3. Find probability:

$$\begin{aligned} P(\bar{x} > 75) &= P(Z > 2) \\ &= 1 - P(Z < 2) \\ &= 1 - 0.9772 \\ &= 0.0228 \end{aligned}$$

 **Answer:** 0.0228 or 2.28%

Interpretation: Only about 2.3% of samples of size 36 will have means above 75.

Example 3: Effect of Sample Size

 **Problem:** A population has $\mu = 100$ and $\sigma = 20$. Compare the standard error for:

a) $n = 25$

b) $n = 100$

 **Solution:**

a) $n = 25$:

$$\sigma_{\bar{x}} = 20 / \sqrt{25} = 20 / 5 = 4$$

b) $n = 100$:

$$\sigma_{\bar{x}} = 20 / \sqrt{100} = 20 / 10 = 2$$

 **Answer:**

- $n = 25$: $SE = 4$
- $n = 100$: $SE = 2$

Interpretation:

- Quadrupling the sample size ($25 \rightarrow 100$) cuts the standard error in half
 - Larger samples give more precise estimates (less variability)
 - To cut SE in half, you need $4 \times$ the sample size
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Example 4: Non-Normal Population

 **Problem:** A population is uniformly distributed (flat, not normal) with $\mu = 50$ and $\sigma = 10$. If we take samples of $n = 40$, can we use the CLT? What is the probability that $\bar{x} < 48$?

 **Solution:**

1. Check if CLT applies:

$$n = 40 \geq 30 \checkmark$$

Even though population is uniform (not normal),
CLT says sampling distribution will be approximately normal

2. Calculate parameters:

$$\mu_{\bar{x}} = 50$$

$$\sigma_{\bar{x}} = 10 / \sqrt{40} = 10 / 6.325 = 1.581$$

3. Calculate z-score:

$$z = (48 - 50) / 1.581$$

$$= -2 / 1.581$$

$$= -1.265$$

4. Find probability:

$$P(\bar{x} < 48) = P(Z < -1.265) \approx 0.1029$$

✓ Answer: 0.1029 or 10.29%

Key Point: Even though the population is uniform (NOT normal), we can use normal distribution methods for the sample mean because $n \geq 30$!

Example 5: Finding Sample Size

 **Problem:** A population has $\sigma = 16$. What sample size is needed so that the standard error is at most 2?

 **Solution:**

1. Set up inequality:

$$\begin{aligned}\sigma_{\bar{x}} &\leq 2 \\ \sigma / \sqrt{n} &\leq 2 \\ 16 / \sqrt{n} &\leq 2\end{aligned}$$

2. Solve for n:

$$\begin{aligned}16 &\leq 2\sqrt{n} \\ 8 &\leq \sqrt{n} \\ 64 &\leq n\end{aligned}$$

✓ **Answer:** Need at least $n = 64$

Interpretation: To achieve a standard error of 2 or less, we need samples of at least 64 observations.

Example 6: Range of Sample Means

 **Problem:** Heights have $\mu = 68$ inches and $\sigma = 4$ inches. For samples of $n = 64$, what range contains the middle 95% of sample means?

 **Solution:**

1. Calculate standard error:

$$\sigma_{\bar{x}} = 4 / \sqrt{64} = 4 / 8 = 0.5$$

2. Use 95% rule (within 2 standard errors):

$$\begin{aligned}\mu_{\bar{x}} \pm 2\sigma_{\bar{x}} &= 68 \pm 2(0.5) \\ &= 68 \pm 1 \\ &= (67, 69)\end{aligned}$$

✓ **Answer:** (67, 69) inches

Interpretation: 95% of sample means (from samples of size 64) will fall between 67 and 69 inches.

Example 7: Comparing Individual vs Sample Mean

 **Problem:** IQ scores have $\mu = 100$ and $\sigma = 15$.

- a) What percentage of individuals have IQ > 115 ?
- b) For samples of $n = 25$, what percentage of sample means are > 115 ?

 **Solution:**

a) Individual scores:

$$z = (115 - 100) / 15 = 1$$

$$P(X > 115) = P(Z > 1) = 1 - 0.8413 = 0.1587 \text{ or } 15.87\%$$

b) Sample means ($n = 25$):

$$\sigma_{\bar{x}} = 15 / \sqrt{25} = 3$$

$$z = (115 - 100) / 3 = 5$$

$$P(\bar{x} > 115) = P(Z > 5) \approx 0.0000003 \text{ (virtually } 0\%)$$

 **Answer:**

- Individual: 15.87%
- Sample mean: ~0%

Interpretation: While individual scores above 115 are common (about 16%), getting a sample average above 115 is extremely rare. Sample means are much less variable than individual scores.

6. Practice Problems

Problem 1

A population has $\mu = 200$ and $\sigma = 30$. For samples of $n = 36$:

- a) What is the mean of the sampling distribution?
- b) What is the standard error?
- c) Describe the shape of the sampling distribution.

Problem 2

SAT scores have $\mu = 1000$ and $\sigma = 200$. What is the probability that a random sample of 64 students has a mean score above 1050?

Problem 3

A population standard deviation is $\sigma = 12$. What sample size is needed for the standard error to be 2 or less?

Problem 4

Waiting times are uniformly distributed with $\mu = 10$ minutes and $\sigma = 5$ minutes. For samples of $n = 50$:

- a) Can we use CLT?
- b) What is $P(\bar{x} < 9)$?

Problem 5

Compare the standard error for $\sigma = 20$ with:

- a) $n = 16$
- b) $n = 64$

What is the effect of quadrupling the sample size?

Answers to Practice Problems

Problem 1

- a) $\mu_{\bar{x}} = \mu = 200$
- b) $\sigma_{\bar{x}} = 30 / \sqrt{36} = 5$
- c) Approximately normal ($n \geq 30$)

Problem 2

$$\sigma_{\bar{x}} = 200 / \sqrt{64} = 25$$

$$z = (1050 - 1000) / 25 = 2$$

$$P(\bar{x} > 1050) = P(Z > 2) = 0.0228 \text{ or } 2.28\%$$

Problem 3

$$\sigma_{\bar{x}} = \sigma / \sqrt{n} \leq 2$$

$$12 / \sqrt{n} \leq 2$$

$$\sqrt{n} \geq 6$$

$$n \geq 36$$

Problem 4

a) Yes, $n = 50 \geq 30$

b) $\sigma_{\bar{x}} = 5 / \sqrt{50} = 0.707$

$$z = (9 - 10) / 0.707 = -1.414$$

$$P(\bar{x} < 9) = P(Z < -1.414) \approx 0.0787 \text{ or } 7.87\%$$

Problem 5

a) $n = 16$: $\sigma_{\bar{x}} = 20 / 4 = 5$

b) $n = 64$: $\sigma_{\bar{x}} = 20 / 8 = 2.5$

Effect: SE is cut in half (50% reduction)

Quick Reference

Sampling Distribution of $\bar{x}$

Mean: $\mu_{\bar{x}} = \mu$

Standard Error: $\sigma_{\bar{x}} = \sigma / \sqrt{n}$

Shape: Approximately normal if $n \geq 30$

Z-Score for Sample Mean

$$z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

Key Insights

- Larger $n \rightarrow$ Smaller $\sigma_{\bar{x}} \rightarrow$ More precise
- To cut SE in half $\rightarrow$ Need 4 $\times$ the sample
- CLT works for ANY population distribution (if n large enough)

The CLT is the foundation for most statistical inference - master it to understand hypothesis testing and confidence intervals!
